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SURFING INSTALLATION

Abstract

The present disclosure relates to a surfing installation for generating a surfable wave, comprising an inclined ramp to the upper end of which water flows via a water inlet. The water is accelerated on the ramp to a first flow speed, the lower end of the ramp leading into a region having a lower second flow speed relative to the first flow speed. A standing wave, in the flow direction, forms in the region due to the water flowing at the first, flow speed striking the water flowing at the second, flow speed. At least two walls may define at least one flow channel and may be arranged in the region of the water inlet and/or of the ramp. The walls are movable transversely to the flow direction of the water to generate at least one traveling wave that travels transversely to the flow direction.

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Background/Summary

RELATED APPLICATIONS [0001] This application is a continuation patent application and claims priority benefit, with regard to U.S. patent application Ser. No. 18/378,908, filed Oct. 11, 2023, (“the '908 Application”). The '908 Application is a continuation patent application and claims priority benefit, with regard to U.S. patent application Ser. No. 16/768,841, filed Jun. 1, 2020, now U.S. Pat. No. 11,788,309, issued Oct. 17, 2023 (“the '309 Patent”). The '309 Patent was filed under 35 U.S.C. § 371, claiming priority benefit, with regard to all common subject matter, of PCT Int'l. App. No. PCT/EP2018/083331, filed Dec. 3, 2018 (“the PCT Application”). The PCT Application claims priority benefit of European Patent Application No. 20170205359, filed Dec. 5, 2017. The above-referenced patent and applications are herein incorporated by reference in their entirety.

[0002] The invention relates to a surfing installation for portable or fixed use.

[0003] A surfing installation of the type in question is known from EP 2 356 298 B1, which can be traced back to the same applicant. In this known surfing installation, the water flows down a ramp and, at the bottom, strikes a standing or more slowly flowing mass of water. The resistance thereof results in the development of what is known as a “standing wave” in the water flowing down the ramp. In this case, a rectangular wave pool also allows for lateral surfing, transversely to the flow direction, but which is limited by the relatively narrow width of the wave pool.

[0004] The object of the invention is that of providing a surfing installation by means of which even better use can be made of a standing wave.

[0005] According to the invention, a moving or traveling “standing” wave is generated in that at least two walls that are arranged before and/or after a ramp or a weir region, so as to be substantially in parallel with the flow direction of the water, and that define at least one flow channel, are movable, by means of at least one drive, transversely to the flow direction of the water flowing down the ramp, and thus generate a moving standing wave in the wave pool, which wave travels transversely to the flow direction, upon striking the standing or more slowly flowing mass of water. The surfer can thus follow the wave that is traveling in the wave pool, transversely to the ramp.

[0006] Since the flow channel is of only a limited width, the energy input required for generating the wave is still no greater, but instead rather less, than in the case of the surfing installation mentioned at the outset. The surfing installation according to the invention thus links as low as possible an energy input and the greatest possible benefit for one or more surfer(s). Irrespective of a reduction in the energy input, a further advantage is a significantly longer surfing distance without turning maneuvers.

[0007] The term “standing wave” means a wave that is stable when viewed in the flow direction of the water, which wave, however, according to the invention, additionally moves further in the wave pool or in a riverbed, as a “traveling wave,” transversely to the flow direction, and thus allows one or more surfer(s) to surf one wave even over a longer distance, without having to constantly make turning movements. A river, a lake or a marine region can also be used instead of a main pool

surrounding the wave pool.

[0008] If the term “movable on the ramp, transversely to the flow direction” is used in this application, in connection with the movement of the walls that define the at least one flow channel, this covers not only movements that occur exactly at a right-angle to the fall line of the ramp, but rather all movements that comprise a component transversely to the ramp. The walls of the at least one flow channel can therefore also be arranged obliquely to the fall line of the ramp.

[0009] If a “ramp” is mentioned in this application, this means both a downwardly sloping surface, downstream of the pumping system in terms of flow, in an artificial surfing installation, and a downwardly sloping region of a riverbed, for example in the region of a weir in a naturally flowing body of water. A ramp can also be formed as an inclined surface or platform that comprises side walls and that floats or is securely anchored in a body of water.

[0010] In an alternative embodiment, a wave pool of an artificial surfing installation is designed in the form of a circular sector having an inscribed angle (central angle) of between 10° and 360° . In this embodiment, the walls that define the flow channel preferably perform a pivoting movement about the center of the circle.

[0011] The movement of the walls defining the flow channel can take place as an alternating reciprocating pivoting movement or, in the case of a wave pool that forms a full circle having an inscribed angle of 360° , also as a continuous rotational movement in just one direction.

[0012] However, in all the embodiments according to the invention, the movement can likewise take place in the form of a translational movement or as a superimposed rotational/translational movement of the flow channel, transversely to the normal flow direction of the water.

[0013] According to an advantageous development, in the case of a surfing installation comprising a wave pool in the form of a circular sector the water inlet is arranged in the region of the center of the circular sector. As a result, the same amount of water is always conducted through the flow channel, irrespective of the current position of the flow channel. In this case, the amount of water delivered depends exclusively on the capacity and the number of the connected pumps.

[0014] According to an alternative embodiment, the at least one flow direction can also be formed by a tube or shaft that is movable transversely to the flow direction in a rotational, translational or superimposed rotational/translational manner, the walls of which then form the walls defining the flow channel.

[0015] According to an advantageous embodiment of the invention, the walls defining a flow channel are connected to the drive by means of a link, and are guided substantially in parallel during a pivoting movement of the flow channel. The speed of pivoting movement can preferably be changed by the drive, as a result of which the requirements can be adjusted to the surfer active in the wave pool in each case.

[0016] It is particularly advantageous for the pumping system for metering the discharge of the water in the water inlet to comprise one or more pumping units, the delivery rate of which can be changed and/or which can be connected individually or in groups. As a result, the amount of water, and thus, in conjunction with a change in the width of the flow channel which is optionally additionally possible, also the height and the speed of the current of water flowing down the ramp, can be changed, as a result of which the height of the traveling wave in the wave pool is also influenced. As a result, the performance level of the surfer in the wave pool in each case, can be taken into account. As a result, the surfing installation according to the invention can always generate a suitable wave, for beginners right up to professional surfers.

[0017] The ramp, in the form of a planar or—in the case of a circular or semicircular wave pool—frustoconical slanted plane is connected, at the upper end thereof, to a flow portion that homogenizes the flow, and is preferably mounted, at the lower end thereof, in a wave pool. The wave pool can alternatively also be formed by a river, a lake or a marine region.

[0018] However, in the surfing installation according to the invention the shape and height of the standing wave is preferably also influenced by the fact that the water flowing down the ramp at a

first flow speed is guided over at least one adjustable guide device that is arranged so as to be spaced apart from the lower end of the ramp, in the flow direction. The guide device can be designed as a fixed assembly, as an assembly that is adjustable with respect to the position and orientation thereof, as an assembly that can be completely sunk, or as a component that is movable along a guideway, the guidance then preferably occurring on a path concentric to the center of the circular sector, in the case of a wave pool that is circular or that is in the shape of a circular sector. The guide device can be arranged as a pivotable floater, directly on the base of the wave pool, or preferably on a pedestal that is raised relative to the base of the wave pool.

[0019] The water level in the wave pool can preferably be set using an adjustable device for drainage regulation. In addition to the second slower flow speed that the water flowing down the ramp at the first, higher speed strikes, said drainage regulation also influences a return flow that flows back to the wave, on the rear face of the wave at the base of the wave pool, and thus also influences the formation of the wave. The adjustable device for drainage regulation is preferably designed such that the water can be let out of the wave pool or forced back therein in a metered manner, in different planes that are arranged one above the other and/or side-by-side. As a result, the return flow acting on the rear face of the surfable wave can be purposefully influenced and adjusted by means of purposeful adjustment of the surface current and undercurrent.

[0020] It is advantageous for a plurality of pumping units, arranged side-by-side, to be arranged upstream of the water inlet, which is also referred to as the flow portion. The flow portion is a relatively large reservoir that homogenizes the amount of water delivered by the different pumping units. As a result, in the case of a lower water requirement for generating a lower wave that is also suitable for beginner surfers, it is possible, for example, for the delivery rate of individual pumps to also be throttled or shut off completely, without this resulting in a different layer thickness of the current of water over the width of the ramp.

[0021] Preferably, each of the plurality of pumping units is provided with one pump, respectively, which pump suctions water from the main pool or from the natural body of water surrounding the surfing installation, and delivers said water to the water inlet (flow portion) upstream of the ramp. In this case, in order to regulate the overall amount of water delivered to the water inlet (flow portion), the delivery rate of the pumps can preferably be regulated and/or the plurality of pumps can be connected and shut off individually or in groups.

[0022] The side walls of the flow portion are preferably adjustably mounted, for the purpose of changing the width and/or the angle of the water supply to the flow channel. This additionally results in further options for influencing the standing wave, for the purpose of regulating the water discharge. It is thus also possible to adjust the height of said wave for different performance levels of the surfers.

[0023] In a mobile surfing installation having a limited circulating amount of water, it is particularly advantageous for the wave pool, the pump chambers, and the collection chamber to be surrounded by a main pool. Separating the higher wave pool from the main pool furthermore guarantees that only a defined mass of water, specifically that located in the wave pool, forms the resistance for the water flowing down from the ramp. This makes the formation of a standing wave substantially more calculable and more clearly reproducible.

[0024] In this case, the separation between the wave pool and the main pool can preferably be regulated by means of the device for drainage regulation. Said device can preferably change both the water storage level in the wave pool and the degree of permeability, for example by means of two grate apertures or aperture plates that are mounted so as to be displaceable relative to one another, or by means of a plurality of flaps that are arranged one above the other and/or side-by-side and can be controlled with respect to the opening width thereof. As a result, the flow speed of the water in the wave pool and the resistance that opposes the water flowing down from the ramp can be adjusted as essential parameters for the form of the shape and size of a standing wave.

[0025] According to a first embodiment, the guide device is formed by at least one guide profile

that can preferably be adjusted not only with regard to the angle of incidence thereof with respect to the flow, but also with regard to the spacing thereof from the lower end of the ramp. It can optionally also be possible for the length of the guide profile to be changed by means of parts that are telescopically displaceable relative to one another. Alternatively or in addition to the guide profiles, the guide devices can also be formed of at least one nozzle strip that is provided with at least one nozzle on which water, delivered under pressure, can act means of at least one pump.

[0026] According to an alternative embodiment, the guide profile is divided into a plurality of mutually adjacent guide profiles, over the width of the wave pool. In this case, in order to influence the wave shape of the wave the plurality of mutually adjacent guide profiles can be arranged over the length and/or over the width of the wave pool so as to be at different positions, transversely or obliquely to the flow. This arrangement at different positions is facilitated in that a plurality of retainers or guideways are provided on the base of the wave pool, for the purpose of fastening or preferably for displaceable mounting of the guide profiles.

[0027] The guide profiles are pivotably mounted in the region of the front edges thereof, and the rear edge thereof is adjustable by means of adjustment mechanics. The adjustment mechanics can be formed for example by a pneumatic cylinder. The pivot bearing in the region of the front edges of the guide profiles is particularly preferably arranged on a pedestal that rises up from the base of the wave pool.

[0028] In a variant, the guide profiles are mounted so as to be freely floating at least in a portion of the pivot region thereof. In this case, a first fold-out movement of the rear edge is achieved using adjustment mechanics, and the remaining part of the fold-out movement occurs automatically owing to the vacuum, caused by the flow, on the upper face of the guide profile. In this case, a boundary, for example in the form of a preferably adjustable band that is arranged between the rear edge of the guide profile and the base of the wave pool, can expediently limit the maximum fold-out angle, such that sudden stalling of the flow and collapse of the wave does not occur. In contrast to the relatively large wave-forming bodies arranged close to the water surface in other known surfing installations (see for example U.S. Pat. No. 6,629,803 B1), the rear end of the guide profile is not located in the region of the standing wave used by the surfers, but instead in a region of the wave pool that is relatively close to the base, under a wave having a significant water layer thickness, such that fall-related injuries cannot be caused by said component.

[0029] In a preferred variant, the guide devices (guide profiles, nozzle strips, pedestal) can preferably be lowered completely into the base of the wave pool, using a hydraulic drive, such that the wave pool can then also be used as a swimming pool, having a completely flat base.

[0030] A further possibility for adjusting the wave can be achieved by the height of an upper bearing in the region of the upper end of the ramp, together with the flow portion, being adjustable by means of adjustment mechanics.

[0031] Finally, further advantageous influencing of the wave is made possible by means of the water level in the wave pool and the outflow quantity of the water out of the wave pool being able to be regulated, at the rear wall thereof that is downstream of the lower end of the ramp, by means of the device for drainage regulation that can be adjusted by adjustment mechanics, from which rear wall the water flows back into the main pool or into the natural body of water surrounding the surfing installation. The return current eddy on the rear face, caused by said backwater, also makes it easier for a surfer who has fallen off the surfboard to stand up in the region of the wave pool downstream of the wave and to climb out of the wave pool.

[0032] An equalizer that consists of a plurality of tubular hollow bodies that are preferably arranged flush side-by-side is preferably arranged upstream of the upper edge of the ramp, the longitudinal axes of which bodies are arranged so as to be substantially perpendicular to the lower end of the ramp. The equalizer causes the water flowing through the flow channel and onto the ramp to be oriented again, irrespective of the particular pivot angle of the flow channel, so as to flow onto the guide profiles (floaters) arranged transversely in the wave pool in a manner

substantially perpendicular thereto. The equalizer additionally achieves eddies and thus additionally enriches the water with air bubbles that ensure greater volume and softer water in the surfable wave. An equalizer of this kind can also be used, in a manner that is advantageous per se, in other known surfing installations, as an additional component.

[0033] According to a further development of the invention that is also advantageous per se, at least one flow divider that is displaceable transversely to the flow direction can be arranged in the wave pool and/or in the region of the ramp. In this case, the at least one flow divider is not only displaceably mounted in the wave pool, but rather can optionally also be used in different positions in the wave pool, and can also be removed therefrom entirely. The wave pool can be divided into one or more portions, and can thus be adjusted to different requirements, using said flow divider or said flow dividers. In the case of a flow divider that is entirely pushed to the side, the entire width of the wave pool is available. If the flow divider is pushed into an intermediate position, in which it divides the wave pool into two portions of the same or different sizes, the rotational, translational or superimposed rotational/translational movement of the flow channel can cover only one portion, and therefore in the case of a low level of operation on the surfing installation at a low delivery rate, partial operation in one of the portions is still possible. It is also possible for different amounts of water and different speeds to act on the different portions, by means of a plurality of flow channels. It is thus possible, for example, for a first region of the wave pool that is divided off by the flow divider and has surfable waves of a lower height to be made available to surfing beginners, while another second region of the wave pool having a higher surfable wave can be made available to advanced surfers.

[0034] Embodiments of the surfing installation according to the invention will be explained in greater detail in the following, with reference to the drawings.

[0035] In the drawings:

Description

[0036] FIG. 1 is a schematic longitudinal section through the surfing installation, when the pumps are shut off;

[0037] FIG. 2 is a schematic longitudinal section through the surfing installation, when the pumps are activated;

[0038] FIG. 3 is a schematic plan view of the surfing installation;

[0039] FIG. 3A is a schematic plan view of a surfing installation comprising a wave pool that widens relative to the water inlet;

[0040] FIG. 4 shows a first arrangement of a plurality of guide devices oriented on a line transversely to the flow;

[0041] FIG. 5 shows a second arrangement of a plurality of guide devices oriented transversely to the flow and so as to be mutually offset;

[0042] FIG. 6 shows a third arrangement of a plurality of guide devices oriented obliquely to the flow;

[0043] FIG. 7 shows a variant in which the guide devices are formed by nozzles or by nozzle strips;

[0044] FIG. 8 is a plan view of a surfing installation according to the invention, in which the wave pool corresponds to a circular sector having a central angle of approximately 90°;

[0045] FIG. 9 is a plan view of a surfing installation according to the invention, in which the wave pool corresponds to a complete circle comprising two pairs of movable walls that are mutually offset by 180° and define two flow channels;

[0046] FIG. 10 is a plan view of an alternative to FIG. 9, comprising a circular wave pool and three pairs of movable walls that are mutually offset by 120° and define three flow channels;

[0047] FIG. 11 is a schematic view of a device according to the invention when used in a natural

river course;

[0048] FIG. **12** is a side view of a guide profile (floater) arranged on a pedestal having a rising front flank;

[0049] FIG. **13** is a longitudinal section through an equalizer that orients the flow of the water, according to the cutting line XIII-XIII in FIG. **3**;

[0050] FIG. **13A** is a longitudinal section through an equalizer which, instead of flow conducting hollow bodies, comprises channels delimited by flow-conducting partition walls;

[0051] FIG. **14** shows a wave pool that is additionally also divided into two portions by means of a flow divider;

[0052] FIG. **14A** shows an alternative to FIG. **14**, in which the flow divider additionally also extends into the region of the ramp; and

[0053] FIG. **15** shows an alternative position to FIG. **14**, in which the flow divider rests on a side wall.

[0054] The drawings in the figures are purely schematic and should in no way be considered as being to scale. The surfing installation **10** shown in the drawings is formed by a tub-shaped main pool **20** that accommodates all the remaining components and retains the water, required for operation of the surfing installation **10**, in a closed circuit. Inside the main pool **20**, a wave pool **30** is arranged on supports **32** so as to be raised with respect to the base of said main pool. In a manner deviating therefrom, in the embodiment shown in FIG. **11** the wave pool **30** and the main pool **20** are formed by a riverbed. In the event of use in a natural body of water, such as a lake or a marine region, the wave pool **30** and main pool **20** can also be formed by said natural body of water.

[0055] Apart from the left-hand part of the main pool **20** shown in FIGS. **1** to **3A**, the wave pool **30** extends over a large part of the length and width of said main pool. However, owing to the base **31** that is raised relative to the main pool **20**, said wave pool is of a small depth. The mass of water located in the wave pool **30** is also correspondingly smaller than the total mass of water of the surfing installation **10**.

[0056] The wave pool **30** is delimited on both sides by two side walls **33**, on the right-hand end face by a rear wall **34**, and on the left-hand end face by a front wall **35**. A device for drainage regulation **36** is arranged in the rear wall **34**, via which device water can flow out of the wave pool **30** and back into the main pool **20** (see FIG. **2**).

[0057] The device for drainage regulation **36** is for example formed by a plurality of flaps **361**, **362**, **363**, **364** that are arranged one above the other and/or optionally also side-by-side, can be pivoted open to a greater or lesser extent or can be completely closed, and that can preferably be adjusted separately from one another by means of at least one adjustment mechanics **37**.

[0058] Instead of the flaps **361**, **362**, **363**, **364**, it is also possible for gratings to be provided, which gratings are displaceable relative to one another and comprise through-openings that can be released or closed to a greater or lesser extent. A further alternative of the device for drainage regulation **36** is a wall part, the height of which can be adjusted, relative to the rear wall **34**, by means of adjustment mechanics **37**, such that the height of the water level in the wave pool **30**, and thus also the overall mass of water present in the wave pool **30**, can be changed.

[0059] For safety reasons, a relatively fine-meshed screen or grating is arranged on the inside of the device for drainage regulation **36**, facing the wave pool **30**, in order to ensure that only water flows from the wave pool **30** back into the main pool **20**, and that no people, carried along by the flow in the wave pool **30** owing to a fall, or objects lost by said people, could be dragged into the main pool **20** and thus into the intake region of the pumping systems **41**.

[0060] A pumping system **40**, denoted overall by **40**, is arranged in the left-hand part of the main pool **20** in FIGS. **1** to **3A**, into which region the wave pool **30** does not extend, owing to the limitation thereof by the front wall **35**, which pumping system is made up, in the embodiment, of a plurality of pumping units **41** that are arranged side-by-side in the transverse direction (see FIGS. **3** and **3A**). A high-capacity volumetric delivery pump **45** is arranged in each of the pumping units **41**,

which pump suctions water from the base region of the main pool **20** and pushes said water upwards. The pumping units **41** are connected, via at least one outlet opening **47**, to a preferably horizontal water inlet **46** which is also referred to as the flow portion and on which the water delivered by the pump **45** is homogenized and subsequently accelerated by side walls **462** that narrow in the flow direction. The side walls **462** may be designed so as to be straight, or, as shown in FIGS. **3** and **3A**, curved. The side walls **462** may consist of a rigid or a resiliently bendable material, and can also comprise a plurality of wall parts that can be displaced telescopically against or into one another.

[0061] In the second part of the flow portion **46**, at least one flow channel **504** is provided, which channel adjoins the side walls **462**, in the flow direction, and is delimited by at least two preferably mutually parallel walls **506** and **508**. The walls **506**, **508** of the flow channel **504** can be pivoted back and forth, continuously or in adjustable intervals, and preferably in a synchronous and rotational manner, by means of at least one drive **5046**, as indicated by the double arrow S (see FIGS. **3**, **3A**, and **11**).

[0062] Instead of a rotational pivoting movement, the walls **506**, **508** can alternatively also perform a translational movement transversely to the flow direction of the water, or, according to a further alternative, a superimposed rotational-translational movement. It is essential to the invention that the water stream delivered by the pumps **45** should be narrowed by the at least one flow channel **504** and moved in rotation and/or translation, back and forth or rotationally, on a circular path, in order to generate a traveling wave **60** transversely to the flow direction of the water.

[0063] A further extremely advantageous detail of the invention, which can also advantageously be implemented separately in other surfing installations, relates to an equalizer **90** that adjoins the movable flow channel **504** in the flow direction. The equalizer **90**, which is formed of a plurality of hollow bodies **94** that are preferably arranged side-by-side and so as to be flush, is arranged upstream of the top end **51** of the ramp **50**, such that the longitudinal axes **95** of the hollow bodies **94** are arranged perpendicularly to the guide profiles (floaters) **55** arranged in the wave pool **30**. As a result, the equalizer **90** causes the water flowing through the flow channel **504** from the ramp **50** to be oriented again, irrespective of the particular pivot angle of the flow channel **504**, so as to flow onto the floaters **54**, **55** or **56** arranged transversely in the wave pool **30** in a manner substantially perpendicular thereto.

[0064] The hollow bodies **94** are preferably designed as tube sections having a round cross section, which are surrounded by a common frame **92** that surrounds the hollow bodies **94** (see FIG. **13**). The hollow bodies **94** can, however, also be formed in different cross sectional shapes, such as square, rectangular, triangular, hexagonal, octagonal or oval. In this case, the intermediate spaces formed between the hollow bodies **94** preferably also function as flow channels of the equalizer **90** that are arranged in parallel with the longitudinal axes **95**.

[0065] For example, in the case of a width of the wave pool **30** of 18 m, the equalizer may be formed by a bundle of tubes **94** that each have a diameter of approximately 100 to 150 mm. The tubes **94** are arranged inside the frame **92** surrounding them so as to be stacked, for example in six to eight layers. In the flow direction, the length of the tubes **94** is advantageously between 200 mm and 500 mm, in particular approximately 300 mm.

[0066] In the case of the wave pool **30** shown in FIGS. **1** to **3A** having a width of 18 m, the side walls **461** preferably narrow the flow portion **46** that adjoins the pumping system **40** in the flow direction, such that the width of the flow channel **504** is approximately one third of the total width, i.e. approximately 6 m.

[0067] The equalizer **90** can advantageously be used, in an advantageous manner, in all the surfing installations **10** shown in the various figures, and furthermore also in already known surfing installations. In the case of a circular wave pool **30** or a wave pool **30** in the shape of a circular sector, the equalizer **90** is adjusted to the curvature thereof or is composed of a plurality of segments that are joined together in a polygonal manner.

[0068] At the side thereof facing the wave pool **30**, the flow portion **46** adjoins the upper end **51** of a ramp **50** that descends obliquely towards the wave pool **30**. The lower end **52** of the ramp **50** plunges directly into the wave pool **30** via the base **31**. The lower end **52** of the ramp **50** is thus in the water of the wave pool **30**. A fixed arrangement of the ramp **50** is preferred. However, according to a variant, said ramp can also be adjustable with respect to the gradient thereof.

[0069] In the variant shown in FIG. **3A**, the wave pool **30** and preferably also the main pool **20** comprise a widening proceeding from the pumping system **40**, owing to oblique side walls **33A**.

[0070] At least one adjustable guide device is arranged at a spacing from the lower end **52** of the ramp **50**, in the flow direction. According to a first embodiment, the guide device is formed by a guide profile **55** that is integral over the entire width thereof and that is arranged on the base **31** of the wave pool **30** or, according to FIG. **12**, on a planar upper face **543** of a pedestal **541** raised relative to the base **31**, such that the pivot angle thereof relative to the flow can be changed by means of adjustment mechanics **58**. Proceeding from the base **31** of the wave pool **30**, the pedestal **541** comprises a rising front flank **542** and a falling rear flank **544**.

[0071] According to FIG. **12**, the adjustment mechanics **58** is designed in the form of a pneumatic cylinder that is arranged between a bearing on the upper face **543** of the pedestal **541**, or, in the absence of a pedestal **541**, close to the base **31** of the wave pool **30**, and a linkage close to the rear edge, on the lower face of the guide profile **55**. However, other units, such as an electric stepper motor that is coupled to a gearing or a toothed rack, can also be used as adjustment mechanics **58**. In a simple variant, the adjustment of the guide profiles **55** can also be achieved manually, in the idle state, when the pumps **45** are shut off (according to FIG. **1**).

[0072] The guide profile **55** can be mounted so as to float freely in the flow, at least in a portion of the pivot region thereof, and/or such that the angle **A** (FIG. **12**) thereof can be mechanically adjusted relative to the flow. In the case of the rear edge of the guide profile **55** being arranged so as to be freely floating at least in the end region, the maximum fold-out angle **A** of said guide profile is for example limited by a band (not shown) that is fastened on the base **31** or on the upper face **543** of the pedestal **541**.

[0073] FIGS. **4** to **6** show three examples for a wave shape **61** that can be achieved by a different arrangement of a plurality of guide profiles **54**, **55**, **56** instead of a single continuous guide profile **55**. In this case, the wave shape **61** in each case schematically reproduces the crest of the emerging wave **60**, in a plan view.

[0074] In FIG. **4**, all three guide profiles **54**, **55**, **56** are mutually parallel on a line, the gaps between the guide profiles **54**, **55** and **56** being as narrow as possible, in reality.

[0075] The forming wave crest accordingly forms in the wave pool **30** perpendicularly to the flow direction or to the side walls **33**.

[0076] In FIG. **5**, the central guide profile **55** is slightly further from the lower end **52** of the ramp **50** than the two outer guide profiles **54** and **56**. As a result, the wave **60** of wave shape **61** assumes the curved wave shape **61'**, comprising a bulge in the flow direction, in the central region. When surfing on a wave of this kind, in each case a relative acceleration occurs when moving towards the middle, and a relative deceleration when moving from the middle towards the outside.

[0077] In FIG. **6** all three guide profile **54**, **55**, **56** are mutually parallel on a line that is at an angle obliquely to the flow direction or to the side walls **33**. During surfing, the wave shape **61''** that follows this slant brings about a relative acceleration when traveling in the direction of the right-hand guide profile **54**, and a relative deceleration when traveling in the opposite direction.

[0078] The examples shown represent only a small selection of the wave shapes **61** that it is possible to create. It is also possible for there to be more or fewer than the three guide profiles **54**, **55**, **56** shown, such that when said profiles are arranged so as to be straight in part and/or oblique in part and/or mutually offset in part, a plurality of wave shapes **61** is possible, beyond the example shown.

[0079] The width of the guide profiles **54**, **55**, **56** can optionally be telescopically adjustable. This

can be achieved in a simple manner by means of an at least partially double-walled design, guide means for a relative displacement of the other wall part being provided on at least one of the wall parts. According to a simple variant, one single continuous guide profile **55** can also extend, in an arcuate arrangement, transversely over the entire wave pool **30**.

[0080] Instead of the guide profiles **54**, **55**, **56**, or in addition thereto, nozzles **71** can also be used as guide devices in order to promote the formation of a standing wave **60**, which nozzles are preferably pivotably arranged either on the guide profiles **54**, **55**, **56** or, as shown in FIG. 7, on nozzle strips **70** that are arranged on the base **31** of the wave pool **30** or alternatively on the pedestal **541**. The nozzles **71** are fed by one or more pumps **72**, the capacity of which is preferably controllable and which suction water out of the wave pool **30** or out of the main pool **20** and deliver it to the nozzles **71** at a high pressure that can be varied by means of the pump capacity. The water jet emerging from the nozzles, which jet comprises a perpendicular component with respect to the main flow in the wave pool **30**, promotes the formation of the wave **60** in a manner similar to the guide profiles **54**, **55**, **56**. The nozzle strips **70** can accordingly be arranged not only, as shown in FIG. 7, on a line transversely to the main flow in the wave pool **30**, but instead can also assume the arrangements, or hybrids thereof, shown in FIGS. 5 and 6. As a result of the pivotable arrangement of the nozzles **71** or the nozzle strips **70** relative to the base **31** of the wave pool, said nozzles or nozzle strips can not only be directed perpendicularly upwards, but instead can also be oriented at any desired angle relative to the main direction, either counter thereto or in the direction thereof.

[0081] In the idle state of the surfing installation **10** according to FIG. 1, a total amount of water fills the main pool **20** and also the wave pool **30** located therein, according to the height of the lowest open flap **361**, **362**, **363** or **364** of the device for drainage regulation **36**. During operation of the surfing installation **10** according to FIG. 2, the pumps **45** start up and deliver water very quickly from the main pool **20**, via the outlet opening **47**, onto the flow portion **46**. From there, the water flows into the flow channel **504**, accelerated by the narrowing due to the walls **462**. In the flow channel **504**, the water is restricted and guided in accordance with the position and orientation of the walls **506**, **508** in each case. Subsequently, the water is directed in the equalizer **90** and enriched with oxygen. From there, the water flows onto the ramp **50** and, with increasing speed, down to the lower end **52** of the ramp **50**. At the lower end **52**, said fast-flowing water meets the water in the wave pool **30** that is calmer or more slowly flowing with respect thereto. The water flowing down the ramp **50** is forced back and formed upwards into a standing wave **60** by the sluggish mass of the almost standing or more slowly flowing water in the wave pool **30**, which wave becomes established in the wave pool **30** as a stationary state and thus forms a laterally traveling surfable wave as a result of the continuous or discontinuous traveling movement of the flow channel **504**.

[0082] In this case, the formation of the standing wave **60** is assisted by the guide profiles **54**, **55**, **56** which function as wave formers. From the wave pool **30**, the water flows back into the main pool **20**, via the device for drainage regulation **36**, and is suctioned again, in said main pool, on the opposing end face of the main pool **20**, by the pumps **45**. The rear wall **34** of the wave pool **30** that accommodates the device for drainage regulation is preferably designed so as to rise obliquely in the flow direction.

[0083] In the simplest case, the mass of water present in the wave pool **30** or in a natural body of water arranged under the ramp **50** spreads out dynamically as a result of the operation of the pumps **45**, such that a physically defined overflow edge to the main pool **20** is not required. In the idle state of the pumps **45** according to FIG. 1, the level of the water in the wave pool **30** corresponds to the level of the water in the main pool **20**. As soon as the pumps **45** suction water out of the main pool **20** and into the wave pool **30**, via the flow portion **46** and the ramps **50**, the level of the water in the main pool **20** drops below the level of the water in the wave pool **30**, such a defined overflow into the main pool **20** occurs in each case. The water is delivered from the main pool **20** into the wave pool **30** in a closed circuit, by means of the pumps **45**, from which wave pool said water flows back into the main pool **20**.

[0084] The invention can be implemented on a mobile installation which has for example approximately the following dimensions and values, all said values being merely by way of example and in no way limiting: The main pool **20** is approximately 25 to 30 m long and approximately 20 m wide. In the idle state of the approximately ten pumps **45**, the water height in the main pool **20** is approximately 1.80 m. The wave pool **30** is approximately 25 m long and approximately 6 to 20 m wide. The base **31** of the wave pool **30** is approximately 1.20 m above the base of the main pool **20**. Therefore, in the idle state, the water height in the wave pool **30** is approximately 0.60 m. During operation, the water height H1 on the flow portion **46** is approximately 1 m, and in the wave pool **30** it is approximately 0.40 to 0.50 m downstream of the wave **60**, and approximately 0.80 m upstream of the wave **60**. The flow channel **504** has a width of approximately 6 to 10 m after the narrowing of the flow portion **46**. The later thickness of the water flowing down in the ramp **50** is approximately between 0.50 m and 1.20 m, depending on the set delivery rate of the pumps **45**. The flow speed of the water at the lower end of the ramp **50** is up to 6 m/s. In accordance with the increasing speed of the water when flowing down the ramp **50**, the layer thickness of said water decreases towards the bottom. In this case, the standing wave **60** reaches a height of up to approximately 1.5 m (measured from the base **31** of the wave pool **30**).

[0085] It is advantageous for the mobility of the surfing installation **10** if at least parts thereof, such as the pumping units **41**, are formed of containers of standard dimensions, a plurality of which are optionally interconnected by means of interposed seals.

[0086] The pumping system **40** has an overall height of approximately 2.80 m. The ten pumping units **41** have a width of approximately 2 m in each case. Each pump **45** delivers approximately 2 m³ water per second. The width of the flow portion **46** tapers, between the entry from the pumping units **41** to the entry into the flow channel **504**, from 20 m to approximately 6.5 to 10 m. The length of the flow portion **46** from the water inlet **47**, including the flow channel **504** and the equalizer **90**, to the upper end **51** of the ramp **50** is approximately 10 to 13 m.

[0087] The ramp is of a length of approximately 4 to 5 m. The height difference between the upper end **51** and the lower end **52** of the ramp **50** is approximately 0.50 m.

[0088] The space in the wave pool **30** in which the standing wave **60** forms adjoins the lower end **52** of the ramp **50**. Said space has a length of approximately 2 m upstream of the guide profiles **54**, **55**, **56**, and a length of approximately 3 to 4 m downstream thereof. The rear wall **34** which rises obliquely to the rear and in which the device for drainage regulation **36** between the wave pool **30** and main pool **20** is arranged adjoins said space, over a length of approximately 3 to 4 m. The guide profiles **54**, **55**, **56** themselves are of a length of approximately 25 cm.

[0089] FIGS. **8** to **13** show various embodiments of a surfing installation **10** according to the invention. In contrast with the surfing installation described above, comprising a rectangular wave pool, the surfing installations according to FIGS. **8** to **11** comprise wave pools **30** that correspond to a circular sector that may comprise a central angle of 10°, up to a full circle of 360°.

[0090] In the case of the surfing installation **10** in FIG. **8**, said central angle is almost 90°. FIGS. **10** and **11** show two surfing installations **10A** and **10B**, the wave pools **30** of which form a full circle in each case.

[0091] The surfing installation **10** or **10A** and **10B** is arranged completely in a main pool **20** which completely surrounds the wave pool **30** and the shape of which is preferably adjusted to the wave pool **30**. The main pool **20** has a radius that is approximately the same size as that of the wave pool **30**.

[0092] The wave pool **30** is separated from the main pool **20** by a device for drainage regulation **36** that can adjust the water level in the wave pool **30** using adjustment mechanics **37**, and can preferably additionally adjust the outflow speed of said water to the main pool **20** and a return flow **62** towards the floater **54**, **55**, **56** to a different level, by means of adjustable flaps **361**, **362**, **363**, **364** or openings below the water line, as a result of which the height and shape of the forming wave **60** can be influenced.

[0093] The water flowing over the device for drainage regulation **36** is received by the main pool **20** and returned to a pumping system **40** below and/or to the side of the wave pool **30**. The pumping system **40** preferably comprises a plurality of pumping units **41**, the pumps **45** of which can be connected and shut off individually or in groups. As a result, the amount of water delivered by the pumping system **40** can be varied over a large range.

[0094] In the case of a surfing installation that is circular or in the shape of a circular segment, according to FIGS. **8** to **10**, the pumping system **40** delivers the water to a water inlet **47** that is arranged centrally in the region of the center of the circle or of the circular sector. From there, the water flows over a flow portion **46** which is horizontal or which descends slightly in the flow direction, and which serves to homogenize the current of water. The at least one flow channel **504** that can be moved in an oscillating or rotational manner is also arranged on the flow portion **46**. The flow channels **504** are each delimited by two side walls **506** and **508**. The lower faces of the walls **506**, **508** adjoin the upper face of the flow portion **46** in a sealed manner. The seal can be achieved for example by means of a rubber strip that is in contact with the upper face of the flow portion **46** and slides on the flow channel in the event of a movement thereof.

[0095] A ramp **50** that descends towards the outside adjoins the flow portion **46**, which ramp is of a frustoconical shape or of a shape corresponding to a sector of a frustum, owing to the circular or circular sector-shaped design of the surfing installation **10**.

[0096] The lower end **52** of the ramp **50** leads into the wave pool **30**. Guide devices in the form of one or more guide profiles **55** and/or in the form of the nozzle strips **70** shown in FIG. **7** are arranged at a spacing from the lower end **52** of the ramp **50**, on a path concentric to the center **48** or on a path extending in a more shallow curve or on a straight or polygonal path. In this case, the guide profiles **55** and/or the nozzle strips **70** are preferably adjusted to the contour of the path.

[0097] The walls **506**, **508** are preferably connected to a support structure or a pivot device which is mounted on a fixed or movable bearing so as to be rotatable about at least one axis of rotation **5048**. The rotation is achieved by means of at least one drive **5046** that is preferably arranged in the region of the axis of rotation **5048**. As shown in FIGS. **3** and **3A**, the walls **506** and **508** that define the flow channel **504** can also be pivoted by two separate drives **5046** that are preferably actuated synchronously. In the embodiment according to FIG. **8**, the walls **506** and **508** can be connected, at the rear end thereof, to a further, semicircular wall **507** which surrounds the water inlet **47** and ensures that the delivered water flows mainly into the flow channel **504**. The walls (**506**, **508**) defining the flow channel (**504**) are connected to at least one drive (**5046**) by a link (**5042**).

[0098] In the embodiment according to FIG. **8**, the flow channel **504** is moved back and forth in an oscillating manner, about the axis of rotation **5048**, by means of the drive **5046**, as shown by the double arrow **S**. As a result of the narrow channeling of the downward flowing water, in the region of the flow channel **504**, when said water strikes the standing or more slowly flowing water in the wave pool **30**, a standing wave **60** forms in each case only in a portion of under the flow channel **504**, which wave travels back and forth together with the pivoting movement of the flow channel **504**.

[0099] A surfer **80** (shown schematically) can thus surf, using his surfboard **82**, on the further traveling wave **60**, along a circular path on a larger distance, transversely through the wave pool **30**. In this case, in the embodiment according to FIG. **8**, said surfer turns shortly before reaching a side wall **33** of the wave pool **30**, and then continues to follow the wave **60** travelling back, which wave is moved in the counter direction by means of the flow channel **504** that pivots back.

[0100] FIG. **9** shows a surfing installation **10A**, in which the wave pool **30** and the surrounding main pool **20** correspond to a full circle. Accordingly, the ramp **50** is also formed as a frustoconical full circle. In the case of the surfing installation **10A**, two flow channels **504** are provided that are arranged so as to be mutually opposing by 180° – ∞ . As a result, two standing waves **60** are formed simultaneously, opposite one another, in the wave pool **30**, and are set into a traveling sideward movement by means of a rotation of the flow channels **504**. As a result, two surfers **80** can surf on

the surfing installation **10A** simultaneously.

[0101] In contrast to the surfing installation **10A**, in the case of the surfing installation **10B** shown in FIG. **10** three flow channels **504** are arranged so as to be mutually offset by 120° in each case. As a result, in the event of a rotation of the flow channels **504** three traveling standing waves **60** are generated simultaneously, which waves can be used by three surfers **80**.

[0102] Depending on the size of the installation, it is also possible that more than three flow channels **504**, for example four or more flow channels **504**, may be provided.

[0103] The surfing installation according to the invention that is shown in FIGS. **8** to **10** allows a surfer **80** or several surfers **80** an entirely new surfing experience, in which a wave **60** can be travelled over a larger distance.

[0104] However, as already shown in connection with FIGS. **1** to **3A**, the surfing installation **10** according to the invention can also be achieved in conjunction with a rectangular wave pool **30**. In the region of the flow portion **46**, further acceleration of the flow speed takes place over limiting walls **462** that taper towards one another, in a funnel-shaped manner, in the flow direction. The walls **506** and **508** that are arranged above the ramp **50** and that define the flow channel **504** are arranged so as to be displaceable transversely to the flow direction, pivotable transversely to the flow direction, or so as to be displaceable and pivotable in combination (translationally and/or rotationally). As a result, the flow of water that is directed into the wave pool **30** via the ramp **50**, by means of the flow channel **504** that is movable transversely to the flow direction, brings about the formation of a changing wave **60** that travels back and forth, in the region of the guide profiles (floaters) **54**, **55**, **56**.

[0105] FIG. **11** shows a variant of a surfing installation according to the invention that is installed in a natural river course. In this case, the installation preferably takes place in the region of a weir that forms a natural ramp **50** owing to the riverbed that falls away in the base region. A flow portion **46** that is located above the weir, i.e. the ramp **50**, is narrowed in a funnel-shaped manner, in the flow direction, by means of two side walls **461**, before transitioning into a flow channel **504** that is delimited by a first wall **506** and a second wall **508** that is spaced apart therefrom.

[0106] The walls **506** and **508** are pivotable, in accordance with the arrow S, by means of a drive **5046**, into a position **506A** and **508A**, respectively, indicated by dashed lines, in which position the flow channel **504A** is positioned obliquely to the flow direction, in the flow portion **46**. In this case, the pivoting movement of the flow channel **504A** takes place continuously or discontinuously. In this case, the pivoting movement can preferably be achieved by a controller, comprising at least one adjustable parameter or at least one program, by means of corresponding actuation of the drives **5046**. This also applies for all the other embodiments in the remaining figures.

[0107] A translational displacement of the flow channel **504** according to the arrow V is also possible, as an alternative to or in a manner superimposed on the pivoting according to the arrow S.

[0108] The narrowing of the flow portion **46** by means of the funnel-shaped walls **461** brings about an acceleration of the flow of water, which acceleration is further increased, in the weir region, by the falling ramp **50**. The back and forth movement of the flow channel **504**, by means of pivoting and/or displacement, generates a standing wave **60**, which travels back and forth and which can be traveled by a surfer **80** using his surfboard **82**, under the ramp **50**, in a region **30** of more slowly flowing water which in this case replaces the wave pool **30** of the other embodiments.

[0109] According to an alternative embodiment, the device comprising the walls **461** and the walls **506** and **508**, can be moved by means of a drive, for example by means of a revolving Bowden cable that is braced above the river course. In the case of this surfing installation which is arranged in a flowing body of water, in contrast to the situation in the case of an artificial surfing installation **10** the energy requirement for the operation of the pumping system **40** is dispensed with entirely.

[0110] A further option for an advantageous embodiment of the surfing installation according to the invention consists in the movement speed of the flow channel **504** being variable. This can also be achieved automatically for example, by means of a camera (not shown) observing the surfer **80**, a

controller evaluating the speed on the basis of the position change thereof, and outputting a corresponding signal to the drive **5046** for pivoting and/or translational displacement of the flow channel **504**.

[0111] Instead of just one flow channel **504**, other embodiments are also possible in which a plurality of flow channels **504** are arranged side-by-side. If said flow channels are charged with partial currents of water of different strengths, a particular shaping of the wave **60** over the width thereof is additionally possible.

[0112] FIG. **12** shows an alternative particularly preferred embodiment of the invention. In contrast with FIGS. **4** to **7**, in this case a continuous guide profile **55**, also referred to as a floater **55**, is arranged in the wave pool **30**. The side view of said floater **55** is shown in FIG. **12**. In this case, the floater **54** is not provided on the base **31** of the wave pool **30**, but instead on a pedestal **541** arranged on the base **31**.

[0113] The pedestal **541** comprises a front flank **542** that is shaped so as to rise in a manner favorable to the flow, a substantially horizontal, plateau-like upper face **543**, and a flank **544** that falls away to the rear. In this case, the pivot bearing **57** of the floater **54** is arranged on the upper edge of the front flank **542** or in the front region of the upper face **543**. The adjustment mechanics **58**, designed by way of example as a pneumatic cylinder, can preferably be arranged so as to be sunk into the pedestal **541** at least in part. The fold-out angle A of the floater **54**, and thus the shape and the height of the wave **60**, can be changed using the adjustment mechanics **58**. The arrangement of the floater **54** shown in FIG. **12**, on a pedestal **541**, can also be used for the floaters **54**, **55**, **56** and also for the nozzle strips **70** shown in the other figures. In an advantageous variant, the pedestal **541**, together with the floaters **54**, **55**, **56** or the nozzle strips **70**, can be sunk completely into the base **31** of the wave pool **30**, such that the wave pool **30** can also be used as a swimming pool having a completely flat base.

[0114] FIG. **13** is a cross-sectional view of an equalizer **90** according to a first embodiment, in which a plurality of tubular hollow bodies **94** are provided within a surrounding frame **92**, in a manner packed so as to rest close together and so as to have parallel longitudinal axes **95**, through which hollow bodies the water flowing down the ramp **50** is once again oriented so as to be in a flow direction extending in parallel with the fall line **53** of the ramp **50** (see FIGS. **3A** and **8**), following deflection by means of the translationally and/or rotatably movable walls **506** and **508**. As a result of the water flowing down perpendicularly striking the stationary or more slowly flowing water, the formation of the standing wave **60** is positively influenced.

[0115] FIG. **13A** is a cross-sectional view of an alternative embodiment of an equalizer **90**. In this case, the water is divided, by means of a plurality of partition walls **96** arranged in parallel in the frame **92**, into individual channels **97** which are preferably of the same size and which, just like the hollow bodies **94** in FIG. **13**, are used for orienting the water in parallel with the longitudinal axis **95** in FIGS. **3A** and **8**. The partition walls **96**, which are preferably formed by separating plates, can also be divided into further channels **97** having preferably rectangular or square cross sections, by means of further partition walls (not shown in FIG. **13A**) that are not arranged in parallel therewith, in particular are arranged so as to be perpendicular thereto.

[0116] The equalizer **90** preferably is of an overall height, from the base of the ramp **50**, of approximately 1 m. In this case, the partition walls **96** are preferably at a spacing of approximately 10 cm to 20 cm.

[0117] FIGS. **14**, **14A** and **15** show a further advantageous feature of the surfing installation **10** according to the invention. In this case, in FIG. **14** at least one flow divider **100**, also referred to as a divider **100**, which is mounted so as to be displaceable transversely to the wave pool **30**, i.e. perpendicularly to the main flow of the water, is arranged in the wave pool **30**, below the ramp **50**. The wave pool **30** is divided into different portions by means of the displaceably mounted flow divider **100**. The wave pool **30** can thus be adjusted to different requirements with respect to the number and the performance level of the surfers. Thus, for example in the case of a relatively low

requirement (for example if only a few beginner level surfers wish to surf), it is possible for a wave **60** traveling back and forth to be supplied only to the part of the wave pool **30** to the left of the flow divider **100** in FIG. **14**, from the flow channel **504** (not shown here) located above the equalizer **90**. In this case, the amount of water delivered by the pumps **45** can be significantly reduced, and thus a significant amount of energy can be saved by throttling the pump or by shutting off a series of pumps **45**. If the number or the demands of the potential users of the surfing installation **10** increase, the region of the wave pool **30** used can be increased again as desired, by displacing the flow divider **100**, or a wave **60** traveling back and forth can act on the larger region of the flow channel **504**, located to the right of the flow divider **100** in FIG. **14**. If the full width of the wave pool **30** is intended to be used, as shown in FIG. **15** the at least one flow divider is moved to a side wall of the wave pool **30** or alternatively also completely removed from the wave pool **30**. [0118] In the variant shown in FIG. **14A**, the flow divider **100** also extends into a region of the ramp **50**, in a shape bent upwards in the region of the lower end **52** of the ramp **50**, in accordance with the gradient of the ramp **50**.

[0119] The at least one flow divider **100** comprises a recess in the lower region thereof, at the height of the guide profile **55**, such that said flow divider can be moved laterally beyond the guide profile **55**.

[0120] As is can be seen from FIGS. **14**, **14A** and **15**, the region to the right and left of the wave pool **30** and beyond the rear wall **34** of the wave pool **30** can be accessed by walkways **110**, **112**, **114**. The walkways **110**, **112**, **114** can be secured by railings (not shown) and reached by steps or ladders on the end face or to the side of the main pool **20**. However, it is also possible for the entire surfing installation **10** to be set into the base, either in part or completely, such that the walkways **110**, **112**, **114** can then be reached at ground level or can be omitted.

[0121] It is optionally possible, according to a further embodiment of the invention, for the walls **506** and **508**, or **506A** and **508A**, which define the at least one flow channel **504**, **504A** and **504B**, to be designed such that they can be raised. The preferably motorized raising of the walls **506** and **508**, or **506A** and **508A**, makes it possible for water to briefly flow over the entire width of the ramp **50** again, as far as the lower end **52** thereof.

TABLE-US-00001 List of reference signs
 10 surfing installation
 5048 axis of rotation
 10A surfing installation
 506 wall (FIG. 11)
 506A wall
 20 main pool
 508 wall
 30 wave pool (or slower flow
 508A wall region)
 51 upper end (of 50)
 31 base (of 30)
 52 lower end (of 50)
 32 support
 53 fall line (of 50)
 33 side wall (of 30)
 54 guide profile (floater)
 34 rear wall
 541 pedestal
 35 front wall
 542 (front) flank (rising)
 36 device for drainage
 543 upper face regulation (of 30 into 20)
 544 (rear) flank (falling)
 361 flap
 55 guide profile (floater)
 362 flap
 56 guide profile (floater)
 363 flap
 57 pivot bearing
 364 flap
 58 adjustment mechanics
 37 adjustment mechanics (for 54-56) (for 36)
 60 wave
 40 pumping system
 61 wave shape
 41 pumping unit
 62 return current eddy
 45 pump
 70 nozzle strip
 46 water inlet (flow portion)
 71 nozzle
 462 side wall (of 46)
 72 pump
 47 water inlet (outlet opening)
 80 surfer of 40 into 46)
 82 surfboard
 48 center
 90 equalizer
 50 ramp
 92 frame (of 90)
 501 side wall (of 50)
 94 hollow body (tube)
 504 flow channel
 95 longitudinal axis
 504A flow channel (of 94; 96, in 504B flow channel parallel with fall
 5046 drive line 53)
 96 partition wall (in 90)
 97 channel
 100 flow divider (divider)
 110 (lateral) walkway
 112 (lateral) walkway
 114 (rear) walkway
 H1 water height (in 46 during operation)
 A pivot angle (of 55)
 S pivot direction (arrow)
 V displacement direction (arrow)

Claims

1. A wave pool for generating surfable waves, comprising: a downwardly sloping surface configured to accelerate flowing water to a first flow speed, the flowing water moving in a first direction; a surfing region that is downstream the downwardly sloping surface and comprises standing water moving at a second flow speed that is lower than the first flow speed, wherein at

least one surfable wave is formed when the flowing water strikes the standing water, the at least one surfable wave travelling in the first direction; and at least one flow channel that comprises at least two moveable walls, wherein: the at least two moveable walls are upstream the downwardly sloping surface, the at least two moveable walls are parallelly moveable in a second direction that is transverse to the first direction, the at least two moveable walls are configured to guide the flowing water to the downwardly sloping surface.

2. The wave pool of claim 1, wherein the at least one flow channel comprises three flow channels and the at least one surfable wave comprises three surfable waves, each of the three surfable waves formed by a respective flow channel of the three flow channels.

3. The wave pool of claim 1, further comprising: at least one adjustable guide that is spaced apart from a lower end of the downwardly sloping surface in a downstream direction.

4. The wave pool of claim 3, wherein the at least one adjustable guide has a telescopically adjustable length.

5. The wave pool of claim 3, wherein the surfing region further comprises: at least one flow divider configured to divide the surfing region into a plurality of subregions.

6. The wave pool of claim 1, further comprising: an equalizer comprising a plurality of tubular bodies arranged side-by-side, wherein the equalizer is arranged upstream of an upper end of the downwardly sloping surface.

7. The wave pool of claim 6, wherein each of the plurality of tubular bodies comprises a longitudinal axis and further comprising: at least one guide, wherein the longitudinal axis is perpendicular to the at least one guide such that the flowing water flows onto the at least one guide perpendicular to the at least one guide.

8. The wave pool of claim 7, wherein the at least one guide comprises at least one of a floater or a nozzle.

9. A wave pool for generating surfable waves, comprising: a downwardly sloping surface configured to accelerate flowing water to a first flow speed, the flowing water moving in a first direction; a surfing region that is configured to receive the flowing water from the downwardly sloping surface and that comprises: standing water moving at a second flow speed less than the first flow speed, wherein at least one surfable wave forms when the flowing water strikes the standing water and travels in the first direction; and at least one flow divider that is displaceable perpendicularly to the first direction.

10. The wave pool of claim 9, wherein the wave pool is circular, and the downwardly sloping surface is frustoconical.

11. The wave pool of claim 9, further comprising: at least one guide that is arranged in the surfing region and that extends across a width of the wave pool.

12. The wave pool of claim 11, further comprising: a base, wherein the at least one guide is arranged on the base or lowerable into the base, thereby presenting a flat base.

13. The wave pool of claim 11, wherein the at least one guide is configured to float in the surfing region and/or has an angle relative to the first direction, wherein the angle is adjustable.

14. The wave pool of claim 11, wherein the at least one guide comprises a plurality of guides arranged in: a first configuration comprising two outer guides and a central guide, the central guide being arranged further downstream from the downwardly sloping surface relative to the two outer guides; or a second configuration comprising three guides arranged parallel on a line that is oblique to the first direction.

15. The wave pool of claim 9, further comprising: at least one flow channel located upstream the downwardly sloping surface and comprising at least two movable walls configured to guide the flowing water so as to form the at least one surfable wave on a side of the at least one flow divider.

16. A method for generating surfable waves, comprising: providing a wave pool, comprising: at least one flow channel having at least two moveable walls; a ramp having an upper end adjacent the at least one flow channel and a lower end; and a surfing region that adjoins the lower end of the

ramp, the surfing region having standing water moving at a first flow speed; delivering flowing water to the at least one flow channel; directing, via the at least two moveable walls, the flowing water to the ramp; accelerating, by the ramp, the flowing water to a second flow speed that is greater than the first flow speed; and forming, via the flowing water striking the standing water, at least one surfable wave in the surfing region.

17. The method of claim 16, wherein the wave pool further comprises at least one flow divider and the method further comprises: disposing the at least one flow divider in the surfing region so as to form a first region having a first surfable wave and a second region having a second surfable wave.

18. The method of claim 17, wherein the first surfable wave comprises a different height than the second surfable wave.

19. The method of claim 16, further comprising: providing at least one pump having an adjustable delivery rate; and controlling at least one of a height or a speed of the at least one surfable wave using the at least one pump in cooperation with the at least two moveable walls.

20. The method of claim 16, wherein the wave pool further comprises at least one flow divider and the method further comprises: providing at least one pump having an adjustable delivery rate; and controlling the adjustable delivery rate based on a width of a portion of the surfing region, the width of the portion defined by a location of the at least one flow divider.
